Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

S.No	Limitation	Impact	Priority to Address
1	Crop recommendations depend on the available dataset.	Prediction accuracy may reduce for unseen conditions or crops.	High
2	Application currently supports only local Flask deployment.	Cannot efficiently support many users at the same time.	Medium
3	Weather information is entered manually by the user.	Incorrect input values may affect prediction accuracy.	High

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users using Flask.	Deploy on a cloud platform to support multiple users.
2	Data Storage	Uses CSV dataset and trained model files.	Store data in a SQL/NoSQL database for better scalability.
3	Performance	Runs on a single local machine.	Optimize the model and improve response time.
4	Security	Basic input validation is implemented.	Add user authentication and secure HTTPS access.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment of the Flask application.	3 Months	Allow multiple users to access the application online.
Phase 2	Integration with live weather API.	6 Months	Improve crop recommendations using real-time weather data.
Phase 3	Support for more crops and advanced Machine Learning models.	9 Months	Increase prediction accuracy and recommendation quality.
Phase 4	Mobile application and multilingual support.	12 Months	Improve accessibility and usability for farmers.